

Raspberry Pi 3 B+ 24x7 Chromium Kiosk

Practical Working + Preparation + KT Guide

Installer version: V5.5.2

Installer package: IoTDashboard_KIOSK_V5.5.2_ARMHF (single-file ARMHF executable)

Primary target: Raspberry Pi 3 Model B+ (1 GB RAM)

Purpose: Run a Chromium dashboard continuously in kiosk/full-screen mode

Document style: Simple operational guide for installation, support and knowledge transfer

SCOPE

This guide explains the practical behavior implemented by the supplied kiosk installer. It does not document the business logic or internal code of the web dashboard itself.

1. What This Project Does

This project converts a Raspberry Pi into a dedicated 24x7 display device. After power-on, the Pi starts an X11 desktop session on tty1 and opens Chromium directly in kiosk mode at the configured dashboard URL.

- No normal desktop login is required for the kiosk.
- systemd owns the full graphical session and restarts it if needed.
- Chromium is supervised separately, so most browser failures are repaired without rebooting the whole Pi.
- The local health monitor checks X11, the visible Chromium window, renderer response, dashboard freshness and memory pressure.
- A separate network guard distinguishes a dashboard/application outage from a real network transport failure.
- Wi-Fi setup can be opened with Ctrl+Alt+W.
- The browser can store the last submitted dashboard login and make one controlled auto-login attempt per tab session.
- Daily maintenance and a periodic browser-only recycle help long-running stability. **Simple Architecture**

Step	Component	Practical Meaning
1	Power ON / Boot	Pi boots to multi-user target.
2	kiosk.service	Runs as the configured normal user on tty1.

3	Preflight	Repairs stale X/Chromium locks, ownership and board-specific X configuration.
4	X11 + Openbox	Starts the display server and lightweight window manager.
5	Chromium	Starts full-screen kiosk mode with RAM cache and low-memory limits.
6	Health + Network Guards	Continuously verify local display/browser health and network/application availability.
7	Targeted Recovery	Browser restart → low-memory recycle → full session restart → guarded reboot only if persistent.

Main Design Goal

24x7	Keep the screen useful for long periods while avoiding unnecessary full -system reboots and unnecessary SD-card writes.
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2. Preparation Before Installation

Prepare	What is Needed	Why
Hardware	Raspberry Pi 3 B+ is the production target. Pi 5 is supported as a test/compatibility target.	Graphics and memory tuning are boardware.
OS	Raspberry Pi OS / Debian-style system with systemd and apt-get.	The installer requires these tools.
User	A normal non-root user, for example: iotdashboard.	Chromium is intentionally not run as root.
Display	HDMI screen connected and powered.	The kiosk starts X11 on display :0 / tty1.
Network	Working Ethernet or Wi-Fi and access to the dashboard URL.	Packages are installed and the dashboard must be reachable.
Remote Support	SSH access is strongly recommended during setup.	The installer takes over tty1 and disables normal tty1 getty/display manager.
Power	Stable PSU suitable for the Pi and display environment.	24x7 use is sensitive to poor power.
URL	Confirm the exact http:// or https:// dashboard URL.	The installer validates URL format and stores it in /etc/default/kiosk.

Before You Run the Installer

- Confirm you can SSH into the Pi.
- Confirm the target kiosk user exists and has a valid home directory.
- Confirm the dashboard URL opens from the Pi/network.
- Record the device name, location, MAC/IP information and support owner.
- Keep the single-file installer `IoTDashboard_KIOSK_V5.5.2_ARMHF` on the Pi and note its exact version.
- Plan one reboot after installation. Automatic reboot is enabled by default.

IMPORTANT

The installer changes the boot path: it sets `multi-user.target`, disables desktop display managers, and masks `getty@tty1` so the kiosk owns `tty1`.

3. Installation – Simple Procedure

Standard Installation

1. Copy the single-file installer `IoTDashboard_KIOSK_V5.5.2_ARMHF` to the Raspberry Pi.
2. On the Pi, make the installer executable:

```
chmod +x IoTDashboard_KIOSK_V5.5.2_ARMHF
```

3. Run the single-file installer with the kiosk user:

```
sudo ./IoTDashboard_KIOSK_V5.5.2_ARMHF iotdashboard
```

4. If a different dashboard URL is required, pass it as the second argument:

```
sudo ./IoTDashboard_KIOSK_V5.5.2_ARMHF iotdashboard https://web.itank.io/login/
```

5. Allow the installation to complete. The executable runs the embedded V5.5.2 installer logic. By default it reboots automatically so graphics, groups, ZRAM and watchdog settings take effect. Single-file means one deployment file; package/network access is still required during installation.

What the Installer Configures

Area	Practical Change
Packages	Installs Chromium, Xorg, Openbox, NetworkManager tools, xdotool/xbindkeys, Python WebSocket support and supporting utilities.
Graphics	Pi 3 B+ uses <code>vc4-fkms-v3d</code> ; Pi 5 uses <code>vc4-kms-v3d</code> . The same SD image can adapt at boot.
Browser Profile	Uses <code>~/.kiosk-chrome</code> and keeps disk/media cache under <code>/run</code> (RAM-backed runtime area).
Memory	Enables ZRAM (50% of RAM when available) and Pi 3 lowmemory sysctl tuning.

Network	Enables NetworkManager and disables Wi-Fi power saving by default.
Logging	Limits system journal growth and retains a controlled amount of logs.
Supervision	Creates kiosk.service, local health timer, network guard, maintenance timer and browser recycle timer.
Backups	Existing managed configuration is copied under /var/backups/kiosk-24x7/<timestamp>.

Expected Result After Reboot

- The dashboard opens automatically in full-screen Chromium.
- Mouse pointer hides when idle.
- Screen blanking and DPMS are disabled.
- No desktop login prompt should take over tty1.

- The health and network supervisors start automatically.

4. Practical Working After Power-On

	Runtime Part	What It Does
A	kiosk.service	systemd starts the kiosk directly as the configured non-root user on tty1.
B	kiosk-preflight.sh	Waits for graphics devices, repairs Pispacific Xorg settings, profile ownership, stale X/Chromium processes and locks.
C	kiosk-session.sh	Starts X11 on display :0. If X exits, it retries with increasing backoff.
D	kiosk-xinit.sh	Starts D-Bus, Openbox, unclutter, Wi-Fi hotkey support and Chromium.
E	Chromium supervisor	If Chromium exits, it is restarted inside the same X session. Repeated rapid crashes can quarantine the browser profile.
F	kiosk-healthcheck.timer	Runs the local health check every 30 seconds by default.
G	kiosk-guard.service	Checks dashboard/network availability every 20 seconds by default and reports to the systemd watchdog.
H	Maintenance / recycle	Daily cleanup plus a browser-only recycle every 24 hours by default.

Browser Behavior

- Kiosk + full-screen mode, using X11.
- Renderer process limit defaults to 2 for the Pi 3 B+ 1 GB profile.
- Browser disk cache defaults to 16 MiB in RAM; media cache defaults to 8 MiB.
- Remote DevTools is bound to 127.0.0.1 on port 9222 for local renderer health checks.
- If Chromium repeatedly crashes quickly, the profile is moved to a .kiosk-chrome.bad.<timestamp> folder and a fresh profile is created.

Dashboard Login Convenience

A small Chromium extension is installed by default for <https://web.itank.io/>. After a user manually submits valid credentials, the extension stores them in Chromium local storage and can fill them later. It performs only one automatic login attempt per tab session to avoid a login loop.

SECURITY

Saved dashboard credentials are stored locally inside the kiosk Chromium profile. Treat the SD card and the kiosk user account as sensitive. Do not place SSH, Wi-Fi or admin passwords in this document.

5. Health Monitoring and Automatic Recovery

Local Health Check

- Verifies Xorg/display :0 is alive and answers xdpinfo.
- Verifies a visible Chromium window exists.
- Uses Chromium DevTools Protocol (CDP) to verify the renderer actually responds.
- Checks dashboard freshness; default maximum age is 900 seconds (15 minutes).
- Checks sustained low memory and swap pressure.
- Default failure threshold is 3 health-check failures.

Recovery Order

Problem	Automatic Action	Why
Chromium/renderer unhealthy	Terminate Chromium and allow the browser supervisor to restart it.	Keeps X/display running if possible.
Low memory sustained	Recycle Chromium before the kernel OOM killer is needed.	Protects systemd, SSH and NetworkManager.
X11/display unhealthy	Restart the complete kiosk.service session.	Rebuilds X + Openbox + browser from a clean preflight.
Local kiosk remains unhealthy	Guarded full Pi reboot after ~10 minutes, with ~6 hour reboot cooldown.	Last-resort local recovery.

Network Guard – Important Difference

KEY POINT	The project deliberately separates a remote dashboard/application outage from a real network transport failure.	
Situation	What the Guard Does	
Dashboard returns HTTP error but network still works	Does NOT restart NetworkManager. It treats this as an application/server issue.	
Local transport is genuinely down	Restarts NetworkManager after the configured failure threshold and cooldown.	
Transport recovers but dashboard is still down	Stops network recovery and waits for the application to recover.	
Transport failure persists	After default 4 NetworkManager recovery attempts and ~1 hour of confirmed transport failure, a guarded reboot may occur. Default reboot cooldown is ~6 hours.	

Dashboard recovers	Chromium is refreshed so the visible page can recover quickly.
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Freshness Monitoring Note

Default freshness mode is 'auto'. It watches meaningful DOM changes and new resource activity. The login path is exempt. If your dashboard can legitimately remain completely static for more than 15 minutes, review the freshness setting during commissioning to avoid unnecessary browser recovery.

TUNING	Available freshness modes in the installer are: off, auto, selector and expression. Use a site -specific selector/expression only when the dashboard team can provide a reliable heartbeat/timestamp.
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6. Day-to-Day Operations

Useful Status Commands

```
systemctl status kiosk.service --no-pager journalctl -u
kiosk.service -b -n 200 --no-pager

systemctl status kiosk-healthcheck.timer --no-pager journalctl -u kiosk-healthcheck.service -b -n 100 --no-pager
journalctl -u kiosk-guard.service -b -n 100 --no-pager systemctl list-timers kiosk-healthcheck.timer kiosk-
maintenance.timer kiosk-browser-recycle.timer
```

Manual Recovery Commands

Need	Command
Restart full kiosk display/browser session	<code>sudo systemctl restart kiosk.service</code>
Restart local networking	<code>sudo systemctl restart NetworkManager.service</code>
Reboot device	<code>sudo reboot</code>
Open Wi-Fi setup at the kiosk	Press Ctrl+Alt+W

GPU Fallback – Only for Real GPU/GL/EGL Failures

```
sudo sed -i 's/^KIOSK_DISABLE_GPU=.*KIOSK_DISABLE_GPU=1/' /etc/default/kiosk sudo systemctl
restart kiosk.service
```

Do not disable GPU as a normal troubleshooting step. Use it only when Chromium logs clearly show GPU/GL/EGL crashes.

Important Paths

Path	Purpose
/etc/default/kiosk	Main kiosk configuration and runtime tuning.

/usr/local/bin/kiosk-*.sh	Generated operational scripts: preflight, session, health, guard, maintenance, recycle and Wi-Fi.
/usr/local/bin/kiosk-cdp-probe.py	Renderer and dashboard freshness probe.
~/kiosk-chrome	Chromium kiosk profile, including local browser data.
~/kiosk-ext	Local dashboard login helper extension when enabled.
/run/kiosk-runtime	Runtime files, browser PID and RAM-backed Chromium cache.

Path	Purpose
/var/lib/kiosk-watchdog	Persistent recovery/reboot guard state.
/var/backups/kiosk-24x7/<timestamp>	Backups created before managed files are changed.

7. Simple Troubleshooting Guide

Symptom	First Check / Meaning	Action
Black screen after boot	Check HDMI/power. SSH in. Check kiosk.service and Xorg-related journal output. Pi 3 uses FKMS by design.	<code>sudo systemctl restart kiosk.service</code>
Chromium not visible	Check kiosk.service logs and healthcheck logs. The browser may be restarting or the profile may be quarantined.	<code>sudo systemctl restart kiosk.service</code>
Dashboard shows error/503	Check whether general network still works. The guard intentionally will not reset networking for a remote application error.	Escalate to dashboard/application owner if network is healthy.
Wi-Fi changed or disconnected	Press Ctrl+Alt+W and use NetworkManager text UI. The browser is refreshed after closing Wi-Fi setup.	Ctrl+Alt+W
Device feels slow / low memory	Check free -m and health logs. ZRAM and low-memory browser recycling are built in.	<code>journalctl -u kiosk-healthcheck.service -b -n 100 --no-pager</code>
Auto-login stopped	The extension allows only one auto-attempt per tab session. Confirm correct dashboard host and manually login once if needed.	Use the on-screen 'Reset saved login' option if credentials changed.
Repeated browser crashes	The supervisor may move the profile to .kiosk-chrome.bad.<timestamp> and start fresh.	Inspect the latest bad profile only for diagnosis; do not blindly restore it.
Nothing recovers	Check service status, local display health and networking. Full reboot is a last resort, not the first action.	<code>sudo reboot</code>

When to Escalate

- Dashboard/application team: HTTP errors while network transport is healthy.
- Network team: repeated confirmed transport failures across the local site/network.
- Hardware team: recurring power loss, undervoltage, SD-card corruption, HDMI issues or device overheating.
- Kiosk/software owner: recurring Xorg, Chromium, profile or GPU failures.

ROLLBACK	The installer creates backups of managed files, but the supplied installer does not provide a one - command uninstall/rollback workflow. Treat rollback as a controlled support activity using the timestamped backup folder or a known-good SD image.
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8. KT (Knowledge Transfer) Section

KT Objective

After KT, the receiving engineer should be able to explain the boot flow, copy and run the single-file installer, install/reinstall the kiosk, check service health, distinguish app outage vs network failure, recover Wi-Fi, restart the kiosk safely, and know when a full reboot is justified.

Suggested 20–30 Minute KT Flow

#	Topic	Demonstration / Explanation
1	Purpose	Explain that this is a dedicated 24x7 Chromium display, not a normal desktop.
2	Boot Flow	Show: systemd → preflight → X11/Openbox → Chromium.
3	Health Flow	Show local health timer, CDP renderer/freshness check and memory check.
4	Network Flow	Explain why HTTP/app outage is not treated as network failure.
5	Hands-on	Run status/journal commands and restart kiosk.service once.
6	Wi-Fi	Demonstrate Ctrl+Alt+W and exit without changing production Wi-Fi unless required.
7	Recovery	Explain browser-only restart, full session restart and guarded reboot order.
8	Ownership	Fill the handover contacts and device details below.

Hands-on KT Checklist

- ☐ Engineer can locate /etc/default/kiosk.
- ☐ Engineer can run systemctl status kiosk.service.
- ☐ Engineer can read kiosk.service journal logs.
- ☐ Engineer can read healthcheck and guard logs.
- ☐ Engineer understands browser failure vs X/display failure.
- ☐ Engineer understands dashboard outage vs real network failure.
- ☐ Engineer can open Wi-Fi setup with Ctrl+Alt+W.
- ☐ Engineer can restart kiosk.service safely.
- ☐ Engineer knows that full reboot is last-resort recovery.
- ☐ Engineer understands locally stored dashboard credentials are sensitive.
- ☐ Engineer knows where backups and quarantined Chromium profiles are stored.

Quick KT Questions

Question	Expected Answer
What service owns the kiosk?	kiosk.service
What should be restarted first for a browser/display problem?	Usually kiosk.service; the automatic system already tries browser-only recovery first.
Does an HTTP 503 mean restart NetworkManager?	No. The guard treats reachable HTTP responses as proof the network path exists.
How do you open Wi-Fi setup?	Ctrl+Alt+W.
What is the default local health interval?	30 seconds.
What is the default browser-only recycle interval?	24 hours.
Where is the main configuration?	/etc/default/kiosk
Where are install backups?	/var/backups/kiosk-24x7/<timestamp>

Support Rule of Thumb

REMEMBER

Check evidence first. Restart the smallest affected layer first. A remote dashboard outage is not automatically a Pi or network fault.

Source reviewed: V5.5.2 kiosk installer logic; deployment package: IoTDashboard_KIOSK_V5.5.2_ARMHF